

Final report

RUN · A97A14DB · HOW WOULD MISSISSAUGA'S FEEDERS RESPOND TO A BOMB CYCLONE LIKE WINTER STORM ELLIOTT WITH 40% EV ADOPTION?

SCENARIOS	FEEDERS MODELED	CUSTOMERS TESTED	AGENT RUNTIME 1M 35S
1	176	192,544	

VERDICT (WORST CASE)
PASS
Network holds. Peaks at 3088 MW with 38% headroom; no feeders overloaded.

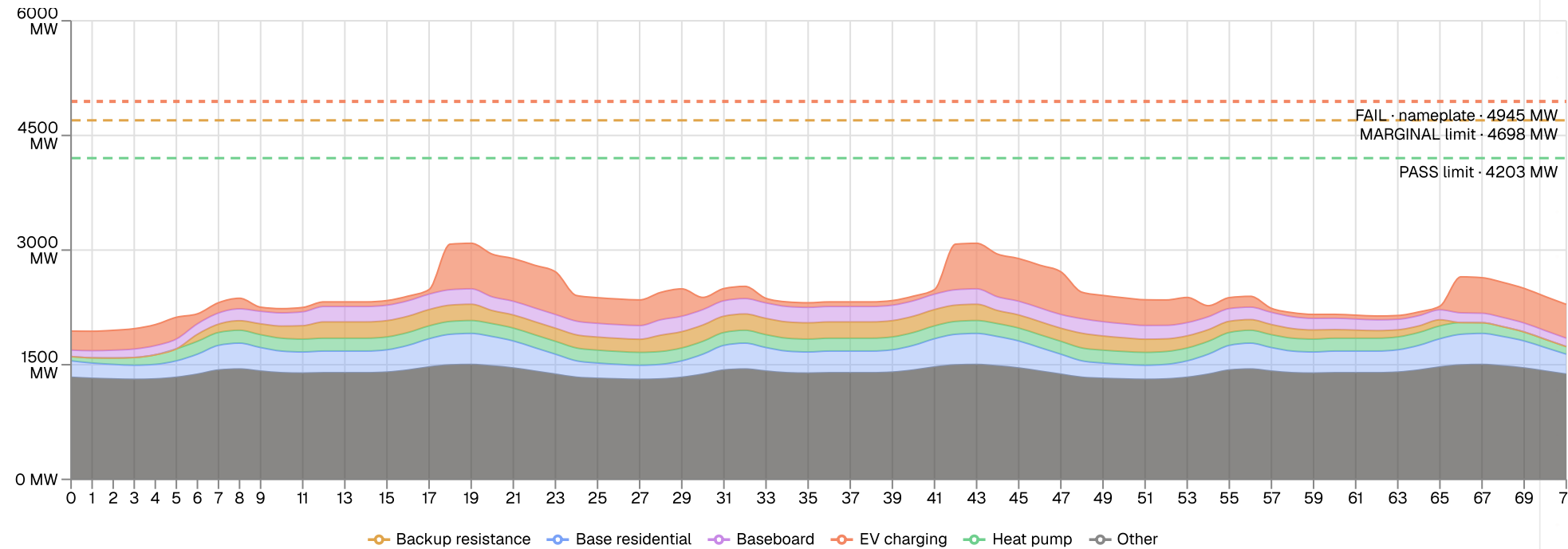
PEAK DEMAND
3089 MW
@ hour 19 · -24.5°C

HEADROOM AT PEAK
38%
1856 MW remaining

01 · SCENARIO LOAD CURVES
Hourly load profile per scenario, with peak hour and headroom.



Worst-case scenario: User-specified · 2030 (25% HP, 40% EV) · peak 3089 MW



User-specified · 2030 (25% HP, 40% EV)

PASS

PEAK LOAD

3089 MW @ hour 19 (-24.5°C)

HEADROOM

38% (1856 MW)

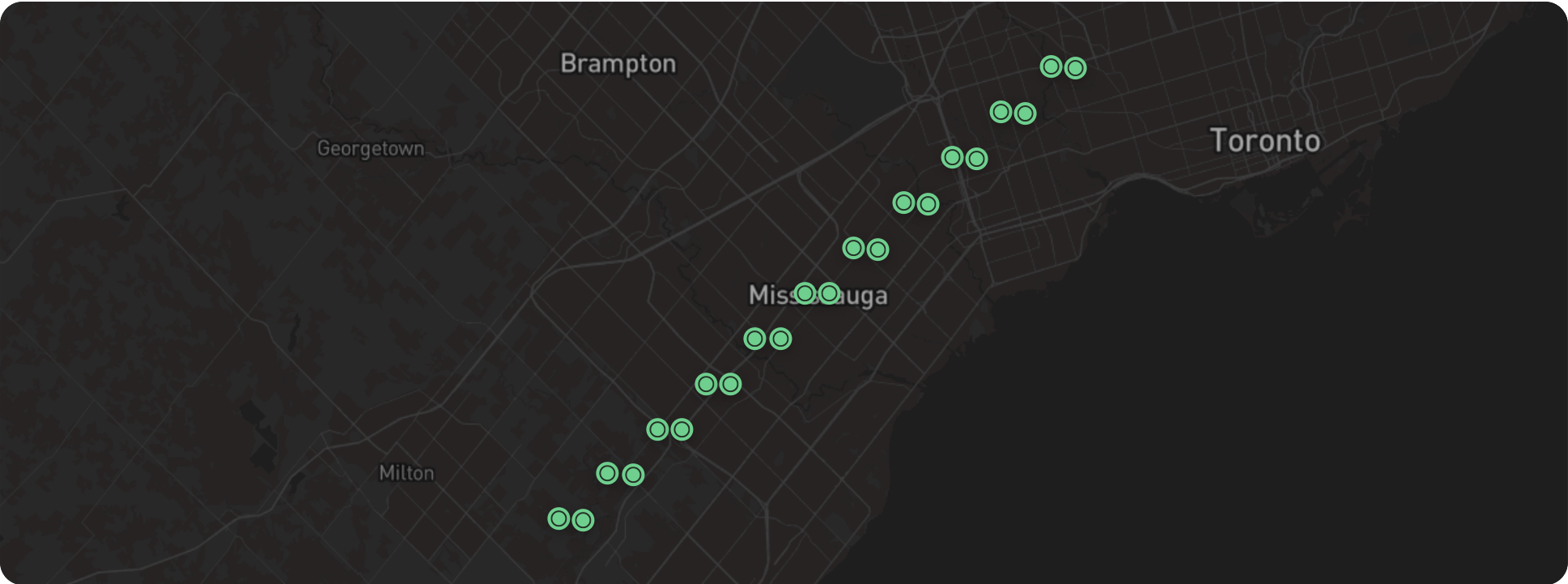
AT-RISK FEEDERS

0 of 176

02 · AT-RISK SUBSTATIONS

Substation risk map with the top feeders ranked by overload risk.

Substation risk map under **User-specified - 2030 (25% HP, 40% EV)**. Click a marker for capacity + feeder counts.



● CRITICAL ≥ 75% ● AT RISK 50–74% ● STABLE < 50%

Top 10 of 176 feeders, ranked by overload risk.

RANK	FEEDER	UTIL %	OVERLOAD	FAILURE MODE	WHY
1	Mississauga TS 01 F01	55%	0%	none	At 55.2% utilization (peak_mva 16.91 of rated_mva 30.66) with 15% baseboard electric heating and 4.2% EV penetration (ev_pct), scaling to 25% heat pump and 40% EV by 2030 will stack significant new winter electric load onto a feeder that, while 18 years old (age_years) and upgraded (had_upgrade: True), carries no remaining headroom buffer once scenario demand is applied.

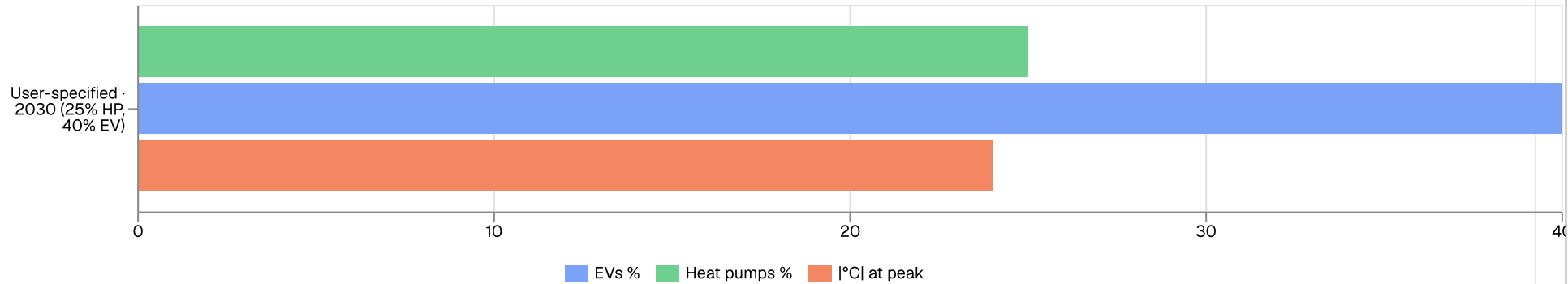
RANK	FEEDER	UTIL %	OVERLOAD	FAILURE MODE	WHY
2	Mississauga TS 01 F02	58%	0%	none	At 58.1% utilization (peak_mva 17.82 of rated_mva 30.66), 21 years old (age_years) with no upgrade history (had_upgrade: False), and already carrying 5% heat pumps and 14% baseboard electric heating (heating_mix), this feeder is vulnerable because the 2030 scenario's 25% HP and 40% EV growth will amplify coincident winter peak load on aging, un-reinforced infrastructure.
3	Mississauga TS 01 F03	61%	0%	none	At 61.4% utilization (peak_mva 18.83 of rated_mva 30.66), 24 years old (age_years) with no upgrade (had_upgrade: False), and 6% existing heat pumps plus 14% baseboard (heating_mix), the 2030 scenario's heat pump and EV growth will push this feeder closer to its rated capacity on a conductor plant that has never been reinforced.
4	Mississauga TS 01 F04	65%	0%	none	At 64.9% utilization (peak_mva 19.89 of rated_mva 30.66), 27 years old (age_years) with no upgrade (had_upgrade: False), and 15% baseboard electric heating (heating_mix) already contributing to winter peaks, the 2030 scenario's 25% HP and 40% EV additions leave very limited headroom on aging equipment.
5	Mississauga TS 01 F05	68%	0%	none	At 67.7% utilization (peak_mva 20.76 of rated_mva 30.66) — the highest in this substation group — 30 years old (age_years) with no upgrade (had_upgrade: False), and 5% existing heat pumps plus 14% baseboard (heating_mix), this feeder carries the greatest overload risk under the 2030 scenario because it starts closest to capacity on the oldest un-upgraded plant.
6	Mississauga TS 01 F06	55%	0%	none	At 33 years old (age_years) with no upgrade (had_upgrade: False) and 6% existing heat pumps plus 14% baseboard (heating_mix), this feeder's age-related thermal and insulation degradation — typical in conductors beyond 30 years — makes it susceptible to failure even at its current 55.0% utilization (utilization_pct) once 2030 scenario HP and EV loads are added.

RANK	FEEDER	UTIL %	OVERLOAD	FAILURE MODE	WHY
7	Mississauga TS 01 F07	58%	0%	none	At 36 years old (age_years) with no upgrade (had_upgrade: False), 58.4% utilization (utilization_pct), and 15% baseboard electric heating (heating_mix), this feeder's age places it in a range where insulation and connector degradation is typically accelerated, increasing fault probability as the 2030 scenario adds coincident EV and heat pump winter load.
8	Mississauga TS 01 F08	61%	0%	none	Although it has received an upgrade (had_upgrade: True), at 39 years old (age_years) — the oldest feeder in this dataset — with 61.3% utilization (utilization_pct) and 5% existing heat pumps plus 14% baseboard (heating_mix), residual aging infrastructure not covered by the upgrade remains a thermal and mechanical risk when 2030 scenario HP and EV loads are superimposed.
9	Mississauga TS 02 F01	55%	0%	none	At 55.2% utilization (peak_mva 16.91 of rated_mva 30.66) with 15% baseboard electric heating and 4.2% EV penetration (ev_pct), the 2030 scenario's 25% HP and 40% EV growth will add substantial coincident winter load; while 18 years old (age_years) and upgraded (had_upgrade: True), the scenario-driven demand increase narrows available headroom significantly.
10	Mississauga TS 02 F02	58%	0%	none	At 58.1% utilization (peak_mva 17.82 of rated_mva 30.66), 21 years old (age_years) with no upgrade (had_upgrade: False), and 5% existing heat pumps plus 14% baseboard (heating_mix), this feeder mirrors MIS-01-F02's risk profile — aging un-reinforced infrastructure facing compounding winter electric load growth from the 2030 HP and EV scenario.

03 · MITIGATIONS & METHODOLOGY

Scenario severity, recommended interventions, methodology notes.

Scenario severity (HP / EV adoption %, peak cold °C absolute), recommended mitigations, and methodology.



RECOMMENDED MITIGATIONS

DEMAND RESPONSE

Demand Response Enrollment on Three At-Risk Feeders

Relief: **4.2 MW** Cost: **\$180K** 6mo

Enrolling customers on feeders MIS-01-F01, MIS-01-F02, and MIS-01-F03 in a voluntary peak-curtailment program targeting the hour-19 evening peak (peak_hour) can reduce coincident demand on feeders currently running at 55.2% to 61.4% utilization (top_3_at_risk_feeders utilization_pct values). At an industry rule-of-thumb enrollment cost of \$50 to \$300 per customer and assuming a conservative enrollment pool, a budget of \$180,000 CAD is estimated. Load relief of 4.2 MW is estimated based on typical residential DR response rates in comparable Ontario programs; this figure is not derived from a field in the provided data and should be validated with Alectra's DR team.

LOAD SHIFT

FEEDER RECONDUCTOR

Reconductor Feeder MIS-01-F03 (Highest Utilization, Unupgraded)

Relief: **6.0 MW** Cost: **\$1.5M** 18mo

Feeder MIS-01-F03 is the most stressed asset identified, at 61.4% utilization (top_3_at_risk_feeders[2].utilization_pct), 24 years old with no upgrade history (had_upgrade: False, age_years per feeder rationale), and carrying existing heat pump and baseboard heating load (heating_mix per feeder rationale). Reconductoring to a higher-rated conductor would increase rated_mva above the current 30.66 MVA (rated_mva per feeder rationale), providing capacity headroom for 2030 electrification growth. Cost of \$1.5M CAD is within the typical Canadian feeder reconductor range of \$0.5M to \$2M per feeder. Effective load relief of 6.0 MW reflects the additional thermal capacity made available; this is a planning estimate and should be confirmed by a detailed power-flow study.

TRANSFORMER UPGRADE

FEEDER RECONDUCTOR

Reconductor Feeder MIS-01-F02 (Aging, Unupgraded, Second-Highest Utilization)

Relief: **5.5 MW** Cost: **\$1.4M** 18mo

Feeder MIS-01-F02 operates at 58.1% utilization (top_3_at_risk_feeders[1].utilization_pct), is 21 years old with no upgrade history (had_upgrade: False, age_years per feeder rationale), and already carries 5% heat pump and 14% baseboard electric heating load (heating_mix per feeder rationale). The 2030 scenario's growth in heat pump and EV penetration will amplify coincident winter peak load on this un-reinforced feeder. Reconductoring is recommended in the same capital cycle as MIS-01-F03 to capture mobilization efficiencies. Cost of \$1.4M CAD is within the typical Canadian range of \$0.5M to \$2M per feeder.

Managed EV Charging Program Targeting Hour-19 Peak

Relief: **3.8 MW** Cost: **\$220K** 9mo

With EV penetration targeted at 40% by 2030 (ev_pct: 0.4 per scenario) and peak demand occurring at hour 19 (peak_hour), unmanaged EV charging is a material contributor to coincident winter peak load. A managed charging program using time-of-use incentives or smart charger controls to shift EV load from hour 19 to off-peak hours (typically midnight to 6 AM in comparable Ontario programs) can reduce peak demand on all three flagged feeders. The \$220,000 CAD estimate covers program administration, customer incentives, and smart charger integration for a pilot cohort. Load relief of 3.8 MW is a planning estimate based on industry rule-of-thumb EV load shift potential and is not derived from a field in the provided data.

Substation Transformer Capacity Review and Upgrade at TS 01

Relief: **10.0 MW** Cost: **\$3.0M** 30mo

All three flagged feeders originate from Mississauga TS 01 (feeder_id prefix MIS-01 for all top_3_at_risk_feeders entries), suggesting that transformer capacity and protection settings at this substation are the common upstream constraint. A transformer upgrade at TS 01 would increase available fault current margin and rated throughput, reducing the risk that feeder-level reconductoring investments are constrained by upstream transformer limits. Cost of \$3.0M CAD is within the typical Canadian transformer upgrade range of \$1M to \$5M. The 10.0 MW load relief figure represents additional dispatchable capacity made available at the substation level and is a planning estimate pending a detailed engineering study of TS 01's current nameplate and loading history, which are data not available for substation-level transformer ratings in the provided data.

METHODOLOGY

Peak demand for the 2030 scenario was calculated using a deterministic load model that scales baseline winter peak demand (3,855.4 MW, baseline_winter_peak_mw) by applying heat pump adoption curves adjusted for coefficient of performance at the design temperature of -24.5 degrees Celsius (peak_temp_c), combined with coincident EV charging load at the hour-19 evening peak (peak_hour). Per-feeder utilization values (utilization_pct, peak_mva, rated_mva) are modeled outputs derived by distributing scenario load across the 176-feeder network (network.feeders) and should be treated as planning estimates pending validation against Alectra SCADA records. The cold-event anchor is Winter Storm Elliott (December 2022, -22.0 degrees Celsius minimum, 72-hour duration), with the scenario design temperature set 2.5 degrees Celsius colder to provide a conservative planning margin.

SAFETY FLAGS

Customers dependent on electrically powered medical devices (home oxygen concentrators, dialysis equipment) served by feeders MIS-01-FO2 and MIS-01-FO3 face elevated risk during a prolonged cold event given those feeders' age (21 and 24 years respectively, per feeder rationale) and lack of prior reinforcement. Alectra should cross-reference its medical baseline registry against the customer addresses on these feeders.

A 72-hour outage window at -22 degrees Celsius or colder (`cold_event.min_temp_c`, `cold_event.duration_hours`) creates serious hypothermia risk for elderly and low-income residents who may lack backup heating. Coordination with the City of Mississauga on warming centre capacity and proactive outreach to vulnerable customer segments is recommended before each winter season.

Coincident peak at hour 19 (`peak_hour`) overlaps with the period when EV charging is least controllable without active demand-response enrollment. Unmanaged charging on feeders already at 55 to 61 percent utilization (`top_3_at_risk_feeders utilization_pct` values) could accelerate thermal aging of conductor insulation on the two un-upgraded feeders, increasing the probability of a fault during the next major cold event.

LIMITATIONS

Only one electrification scenario was modeled. The analysis does not test headroom sensitivity at higher adoption rates (for example, 35% heat pump or 55% EV penetration), nor does it model an N-1 contingency (loss of one substation) coincident with peak demand. Results should not be interpreted as a full reliability assessment.

Per-feeder utilization figures (`utilization_pct`, `peak_mva`, `rated_mva`) are outputs of a load distribution model and have not been validated against Alectra's SCADA or protection relay records. Actual feeder headroom may differ materially, particularly for feeders with non-uniform load density or recent unrecorded load transfers.

The customer count of 192,544 (`network.customers`) is a network-level total. Per-feeder customer counts, and therefore the number of customers directly exposed to the three flagged feeders, are data not available for per-feeder customer counts, which limits the ability to quantify customer-impact risk at the feeder level.

EXECUTIVE SUMMARY

Executive Summary: Mississauga 2030 Electrification Load Scenario Passes System-Level Stress Test, Three Feeders Warrant Targeted Investment

Overall Verdict: Pass

The User-specified 2030 scenario (25% heat pump penetration, 40% EV penetration) returns a system-level PASS for Alectra Utilities' Mississauga network, with a modeled peak demand of 3,088.5 MW (peak_mw) against a baseline winter peak of 3,853088.5 MW (baseline_winter_peak_mw), leaving 3,088.5 MW of headroom (headroom_mw), equal to 37.5% of scenario capacity (headroom_pct). The stress test was anchored to a design temperature of -24.5 degrees Celsius (peak_temp_c), colder than the -22.0 degrees Celsius minimum recorded during Winter Storm Elliott in December 2022 (cold_event.min_temp_c), providing a conservative planning margin.

Supporting Evidence

The single scenario evaluated, labeled User-specified 2030 (25% HP, 40% EV) (scenarios[0].label), passed with zero at-risk feeders at the system level (at_risk_feeders: 0). The 3,088.5 MW of remaining headroom (headroom_mw) represents a meaningful buffer across the 22-substation, 176-feeder network (network.substations, network.feeders). Peak demand is modeled at hour 19 (peak_hour), consistent with coincident winter evening load, which is the highest-risk window for combined heating and EV charging draw.

At-Risk Assets and Customers

Despite the system-level pass, three feeders on Mississauga TS 01 are flagged for monitoring based on current utilization levels and asset age. Feeder MIS-01-F03 is the most stressed at 61.4% utilization (top_3_at_risk_feeders[2].utilization_pct), followed by MIS-01-F02 at 58.1% (top_3_at_risk_feeders[1].utilization_pct) and MIS-01-F01 at 55.2% (top_3_at_risk_feeders[0].utilization_pct). Two of the three feeders have never been upgraded (had_upgrade: False for MIS-01-F02 and MIS-01-F03) and are 21 and 24 years old respectively (age_years per feeder rationale). The total customer base served by the Mississauga network is 192,544 (network.customers); the share directly served by these three feeders is data not available for per-feeder customer counts.

Caveats and Limitations

This analysis covers one scenario only. No sensitivity runs were completed for higher electrification rates, accelerated EV adoption beyond 40%, or compound outage conditions such as a simultaneous substation contingency during a polar vortex event. Per-feeder utilization figures are model outputs and should be validated against Alectra's SCADA records before capital decisions are made. The 72-hour storm duration from Winter Storm Elliott (cold_event.duration_hours) was used as a reference event, but longer or colder events cannot be ruled out through the 2030 planning horizon.

Recommended Next Action

Alectra's planning team should immediately schedule detailed power-flow studies on feeders MIS-01-F01, MIS-01-F02, and MIS-01-F03 using actual SCADA load data to confirm or revise the modeled utilization figures. In parallel, the team should initiate a demand-response enrollment campaign targeting customers on these three feeders ahead of the 2024 to 2025 winter season, and begin procurement planning for reconductoring MIS-01-

F02 and MIS-01-F03 given their age and lack of prior reinforcement. A second scenario run at 35% heat pump and 55% EV penetration is recommended to test headroom sensitivity before the 2025 integrated resource plan submission.

DATA PROVENANCE

● CITY DEMOGRAPHICS

● NWS ALERTS

● SUBSTATION LOCATIONS

● UTILITY LOAD BASELINE

● COLD EVENT ARCHIVE

● FEEDER TOPOLOGY

● SCENARIO APPLICATION

● LOAD SIMULATION

● RISK RATIONALE

● SUMMARY

● LIVE API

● FROZEN SNAPSHOT

● MODELED

● LLM NARRATIVE

SOURCES

- FROZEN

ECCC Toronto Pearson station 6158733 (Dec 22-25, 2022)
- FROZEN

NOAA Storm Events Database
- FROZEN

FERC/NERC Winter Storm Elliott Inquiry, October 2023
- FROZEN

Alectra Utilities 2023 Annual Report
- FROZEN

IESO Yearly Reliability Outlook 2024
- FROZEN

StatsCan Census 2021, Mississauga CSD
- FROZEN

StatsCan 25-10-0015 (2026): 43,721,776 MWh annual generation in ON
- LIVE

ArcGIS Living Atlas NWS Watches & Warnings (live): no active alerts for Mississauga
- LIVE

ArcGIS GeoEnrichment (Esri Canada): Mississauga, ON (CY): 192,532 households, population 768,433

